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# Well Leak Assessment After Shallow Water Well Blowout: A Case Study in West Kuwait

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## Abstract

A gas blowout event occurred in a shallow water well, conventionally used as a brackish water source for drilling rigs, in the Minagish Field located in West Kuwait on March 2023. After 16 months of continuous production, the well was effectively controlled by drilling vent wells. A study was undertaken to assess the specific source reservoir gas leak path(s) from the deeper Minagish formations to the shallow water formation, assessing geological leak paths, well integrity and narrowing down the possible problem well(s) from the ~600 wellbores in the field.

A review and quality control of a very large volume of subsurface and wells data was undertaken by an integrated team of discipline subject matter experts. Production Chemistry fingerprint work identified the primary source of the gas to narrow the focus of the review. In the water formation, a characterisation study was performed using mud loss data (presence of vugs, high permeability channels, fracture channels, etc..), to highlight the main karst distribution both areally and vertically which was then mapped. The maps provided both guidance on leak flow paths and possible locations of future vent wells if required.

The study concluded that the gas leak was primarily due to internal and external casing corrosion in a number of wells, exacerbated by the challenging cement placement in the loss prone shallower hole sections. By defining and adopting a range of leak evaluation criteria, the ~600 suspect wellbores were reduced to 24 possible potential leak wells connecting the Minagish reservoir to the water formation, two of them classified as high risk and another one as medium. In support of this well leak path conclusion, this study also concluded the leak is "unlikely a geological leak path", which might require a leak path from the Minagish formation through multiple reservoirs and seals. There was firm geomechanical and chemical evidence supporting this conclusion, including:

- Several seals that were present
- Over-pressured or depleted zones that could act as a natural flow barrier
- No evidence of deeper faults reaching the water formation

- No data supporting fracturing caprock or fault re-activation in the overburden
- Direct leak path to the water formation - the blowout hydrocarbons comprised of native Minagish reservoir oil and injection gas, with no evidence of interaction with fluids from other reservoirs.

The integrated multidisciplined nature of the study and the application of chemical analysis to narrow the well suspects was the key to evaluating the most likely mechanism and specific wells of concern.

## Introduction

Located in the south of Kuwait, the mature Minagish (MN) field, in production since the 1960s, is formed by a stacked mix of clastic, carbonate and fractured carbonate reservoirs (Figure 1) (Youash and Mukhopadhyay, 1982; Al-Ajmi et al., 2001; Al-Mutairi, et al. 2001). In total there are approximately 440 oil development and water wells in a congested surface location (over 600 wellbores). The majority of production comes from the Minagish oolite limestone which has excellent reservoir properties, including, 15-25% porosity, 300mD-1D permeability, 32° API oil and 3200-3500 psi reservoir pressure. Brackish water (< 3000 mg/L) is also produced from the overlying shallow water formation (not shown in Figure 1, please see the stratigraphic column in Figure 3 for reference) for industrial purposes, including for drilling fluids.

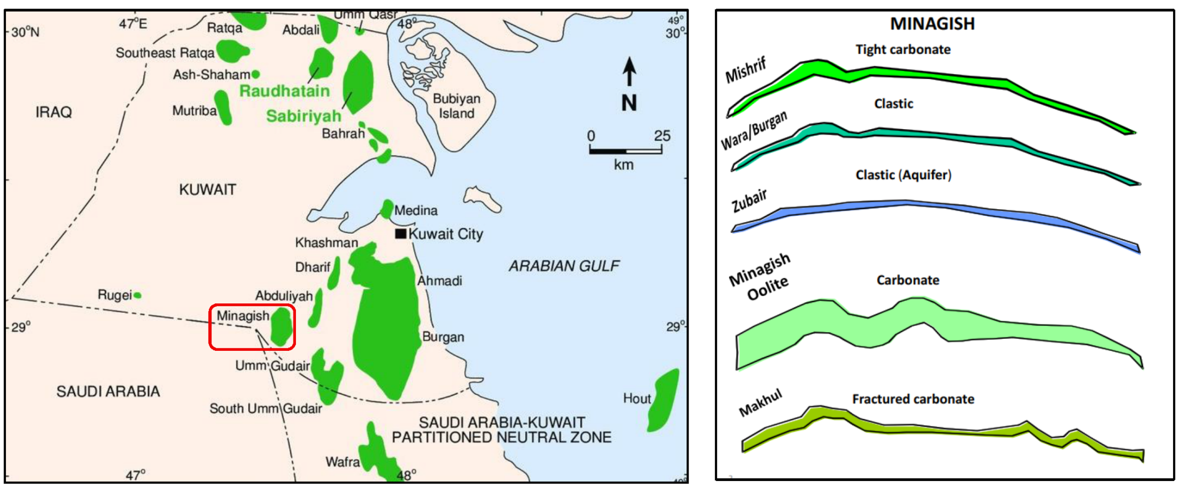



**Figure 1—Minagish oil field location map & high-level stratigraphy schematic.**

In March 2023, a water well experienced a blowout. While dominated by gas, the leak also brought with it significant volumes of crude oil and water. While well control was quickly regained, despite extensive investigation and successful application of relief wells, the source of blowout fluids and leak paths to incident well were not fully understood.

## Aims & Objectives

The overriding concerns were to identify the cause of the leak and put in place mitigations to ensure the hydrocarbon influx (and associated pressurization) into the shallow water formation was halted, with this returned safely and environmentally soundly to water production only. The aim of the study therefore was to determine the source of leak fluids and pathway to surface, facilitating remediation and prevention of further such incidents, in line with KOC's objectives.

## Methodology

To achieve study objectives, the fluids were chemically fingerprinted to determine source, the potential for geological leak pathways was assessed, along with well integrity and associated possible leak routes via the ~600 wellbores in the field. This led to possible well repair candidates and also focused data acquisition to reduce leak path uncertainty. Geological paths were assessed by reviewing all permeable formations and by characterizing the shallow water bearing formation. To support further reservoir integrity, possible future vent well locations were also identified.

The problem as presented had multiple facets, that were all interlinked. Working in isolation on one or two of them risked missing an important part of the jigsaw puzzle that could unlock root causes. Bringing together the technical parts of this problem was key to understanding the potential leak paths and ultimately building confidence as to the correct remedial actions. Each technical discipline has its own clear goals, workflows and outcomes, however it was the integration of those that led to the overall conclusions reached and success of the project in achieving its aims.

Figure 2 shows an infographic describing the integration of various disciplines involved in this project. The goals, workflows and outcomes of these are discussed individually in the next section.


**Figure 2—Infographic describing the integration of various disciplines involved in this project.**

## Multidisciplinary Workflows & Results

### Geology

The Geology element of the project was focused on constructing a detailed stratigraphic framework for the field, identifying zones of flow potential, and developing a deeper understanding of the shallow formations, including establishing any horizontal leak paths. The primary aim was to establish whether a geological leak path was plausible as a route for hydrocarbon migration, and as a secondary outcome a deeper exploration of the shallow water bearing formation and its karst structure.

***Geology Workflow.*** The stratigraphic framework in the Minagish field study was built using a combination of geological, petrophysical, and subsurface data sources, alongside methodical reinterpretation and regional stratigraphic correlation techniques. Various data sources (daily drilling reports, mudlogs, well schematics, Petrel™ projects, core, well logs and broader background from regional papers and stratigraphic charts) were used to update supplied Petrel™ models. Where possible, lithology was confirmed from mudlogs or core descriptions, with key formations characterized based on dominant lithofacies, depositional environment and permeability indicators. Petrophysical analysis was conducted on a number of wells to validate porosity, net-to-gross (NTG) and fluid content. Correlation of permeable units with dynamic data (e.g. losses) helped redefine and validate formation boundaries, especially in transition zones.

***Geology Results.*** A fieldwide stratigraphic framework was developed, shown in Figure 3. This covered the entire stratigraphic column from the Kuwait Group (Tertiary) down to Minjur (Triassic). This supported the well review element to identify geological leak paths and any natural seals.

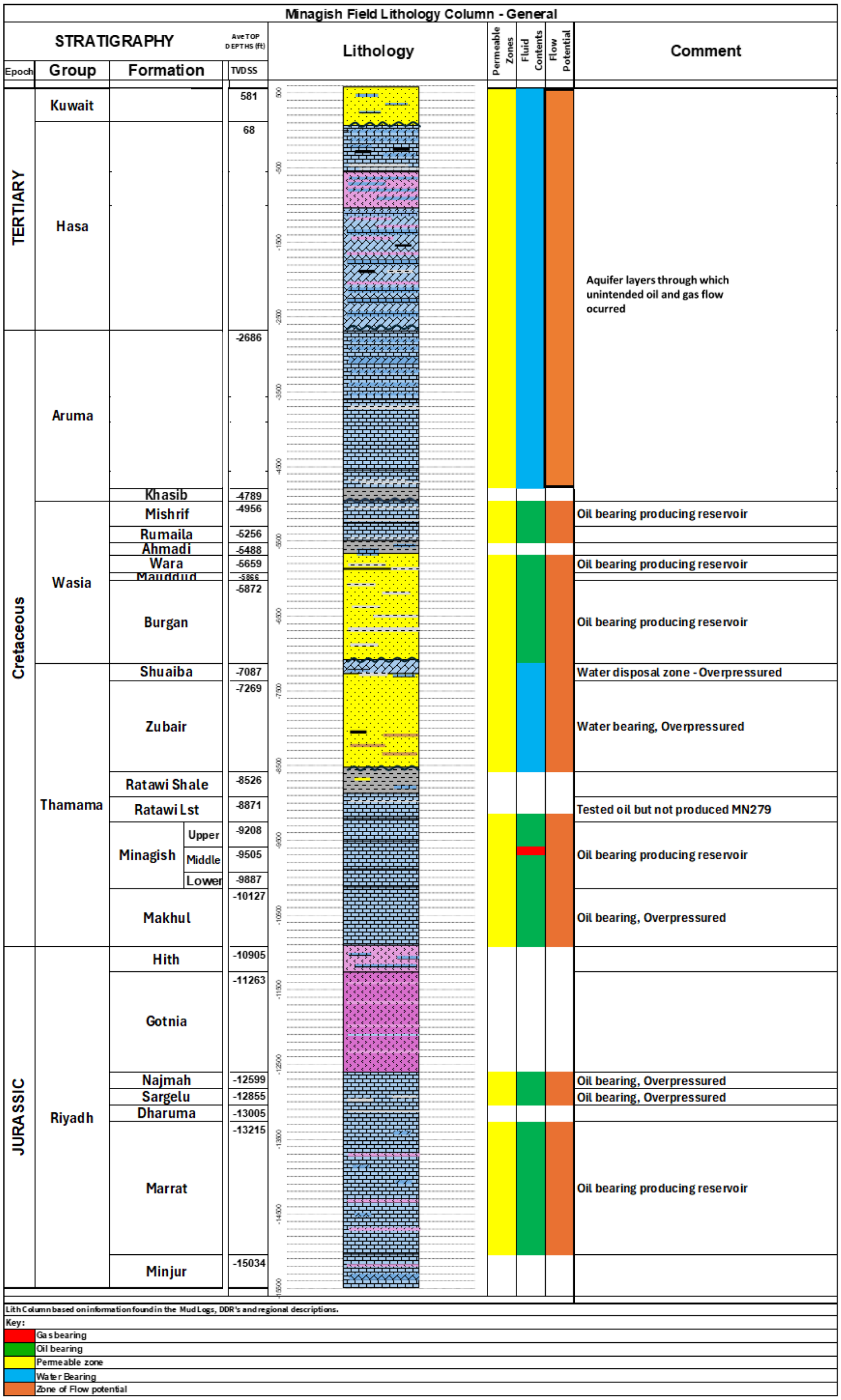



**Figure 3—Schematic Illustration stratigraphic framework development covering the entire stratigraphic column from the Kuwait Group (Tertiary) down to Minjur (Triassic), identifying key features including permeability, fluid fill and established cap rocks.**

For each permeable formation, different anomaly maps were created (80 in total) and an example is shown in Figure 4. These assisted with the identification of potential fluid migration pathways.

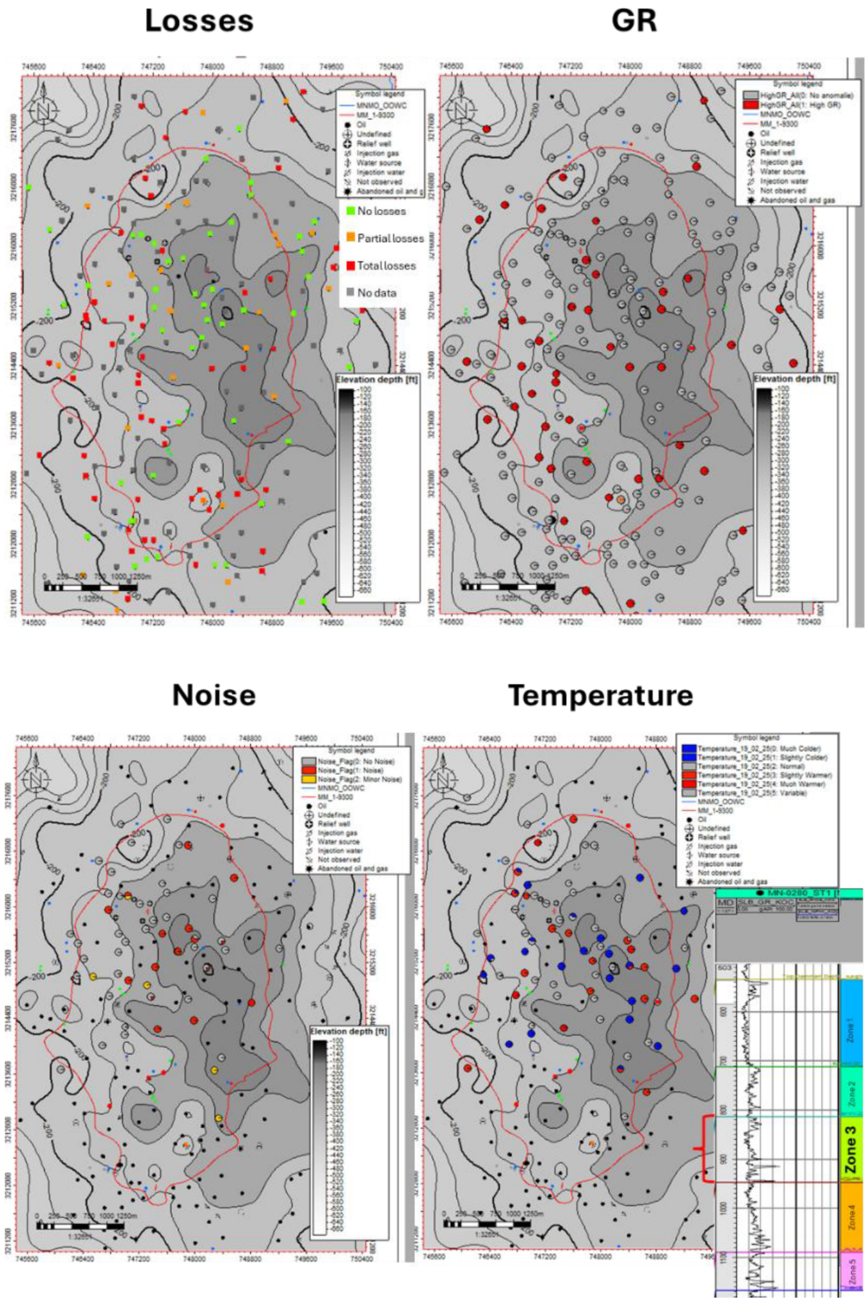



**Figure 4—Example of various surveillance data shown on a structure map to highlight areas of interest.**

Three shales were identified between the middle Minagish reservoir and the shallow water formation which could be considered barriers to vertical flow, which helped to support the well review elements for potential leak candidates. Everything above the Kashib was considered permeable, and as such the leak path only had to reach one of these formations to ultimately get to the water formation, and therefore to surface.

The shallow water formation itself is a complex Karst system (Al-Awadi et al., 1998a,b; Misak and Hussain, 2023; Schehab et al., 2023), so extensive modelling was undertaken using observed well mud loss events to map the potential Karst thickness (Figure 4 and Figure 5) and, hence, to show potential pathways within sublayers of the formation between wells at quite some distance from the blow out event. These pathways could be very small and wells geographically close to each other could find themselves not equally connected to these pathways, such as the first relief well which failed to find gas.

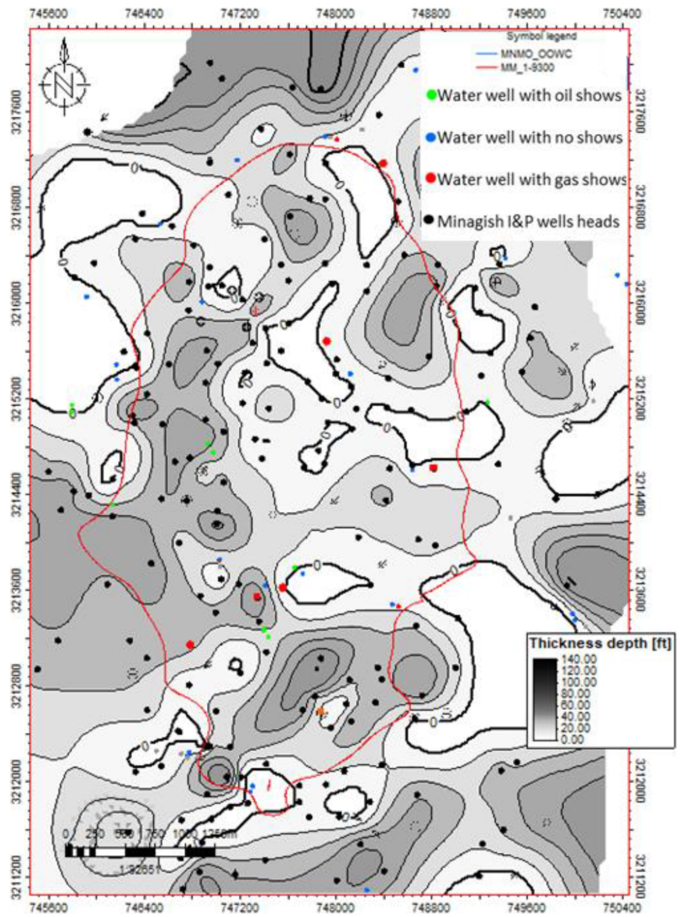



**Figure 5—Modelled Karst system within the shallow water Fm. based on developing a detailed geological understanding of the structure and detailed review of well loss events which are linked to entering the Karst. Potential Karst thickness is mapped with thicker zones appearing darkest and thinner zones lighter, areas of no losses and hence no interpreted Karst are white.**

Figure 5 shows the modelled Karst system within the shallow water formation. This model was developed independent of known well events (gas blowouts) or water well sampling, however when this data was combined, the resultant map showed a good match between thicker areas of Karst and event wells. This mapping can be used to inform potential risks while drilling for shallow gas pockets in the shallow water formation and to identify potential locations for targeted vent well drilling.

## Geophysics

There was an initial suspicion that gas may be leaking via a series of reactivated faults and fracture systems. The focus of the Geophysics component was therefore to map potential fracture and damage zones at the main sealing intervals with the aim of proving / disproving this. Once addressed, subsequent work was recommended to focus on alternative monitoring methods and the potential for high resolution seismic acquisition to image the shallow water formation karst systems not covered by the existing seismic volumes.

***Geophysics Workflow.*** A data review and QA/QC of the supplied seismic volumes was undertaken. From this, a PreSTM 3D volume was found to be the best of the available datasets and was used for amplitude and spatial attribute analysis interpretation. Potential fracture and damage zones were mapped at the main sealing intervals and the results shared with other disciplines in a suitable format.

A range of spatial attributes and parameters were tested (similarity, semblance, entropy and ESP) with similarity proving the most suitable in this context for highlighting broad damage zones and possible fracture corridors. The Similarity volume was binned appropriately to provide a simplified qualitative measure of seismic variability that could be incorporated into Well and Geomechanics studies (Figure 6 and Figure 7).

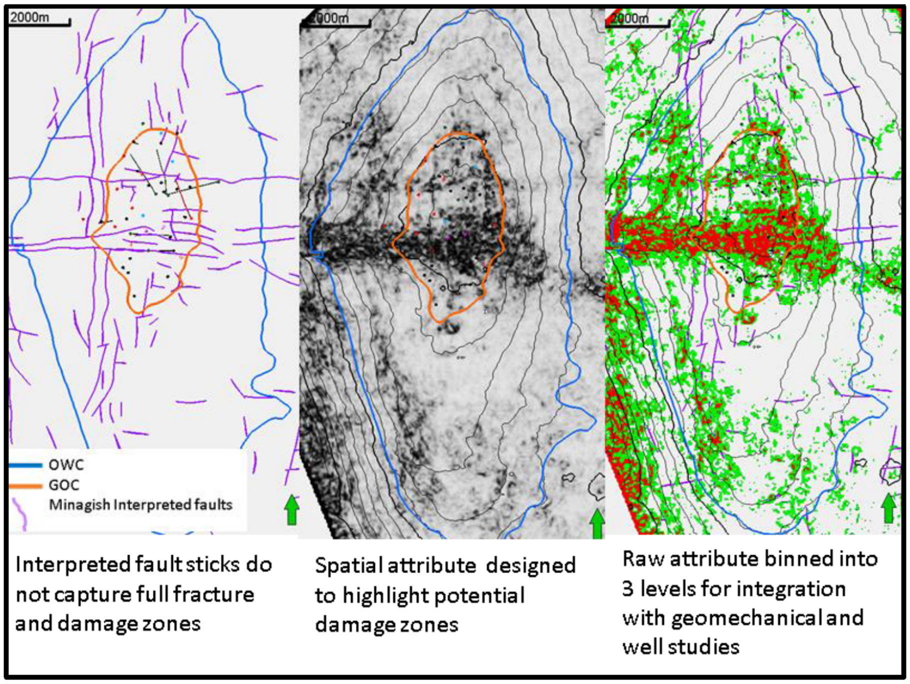



**Figure 6—Seismic variation at Top Ratawi showing interpreted faults, the use of special attribute to highlight potential damage zones, and a categorization of the raw attribute for the broader team to use in their analyses.**

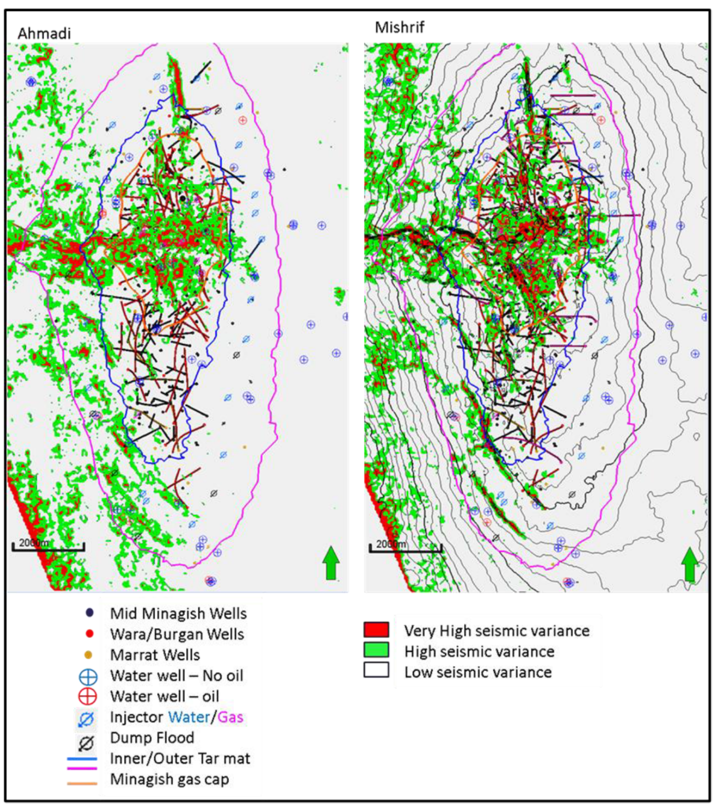



**Figure 7—Seismic variability at the Ahmadi and Mishrif formations.**

***Geophysics Results and Conclusions.*** There are clear fractures and faults present which have been included in the overall analysis. From a geophysical aspect, there are no amplitude anomalies at any reservoir levels that could be linked to hydrocarbon presence. No vertically continuous features were found linking the Ratawi formation to the top of the available data. The seismic data has a highly discontinuous character in the Hartha, Sadi and Tayarat formations consistent with the presence of karst as suggested by the geological work.

An investigation of alternative monitoring and imaging tools applicable to gas leaks found that high resolution shallow seismic and satellite methane detection showed the most promise.

## Reservoir Engineering

The Reservoir Engineering elements of this project focused on a review of the supplied dynamic models and history matches, pressure data analysis to support Geomechanics, gas cap behavior assessment and a basic material balance analysis to investigate the magnitude of the gas leak. The aim was to explore the pressure characteristics of the reservoir over time, and to more deeply understand where fluids could have moved to within the various reservoir structures.

***Reservoir Engineering Workflow.*** The supplied dynamic models were reviewed and a detailed statistical error analysis performed with the intent of determining the wells with the largest history match error. These could then be further investigated at a well level to see if they were contributing to the vertical leak. Maximum and minimum pressures seen across the field were calculated and mapped to support the Geomechanics team. A comparison between the logged (Pulse Neutron Capture [PNC]) and modelled gas-oil contacts (GOC's) was also performed, to support the Wells review team, as any leaking well would have to be within the gas cap zone of the reservoir due to the chemical fingerprinting. It was hoped that any discrepancy between logged and simulated contacts may have led to refining the location of the leak, but this was not possible due to the GOC mismatch in the model.

A material balance model was developed in an attempt to detect and quantify a leak from the Minagish reservoir. Any historical pressure decline not explained by production/injection could be an indication of the leak. This was performed by comparing observed well pressures, dynamic model pressures and material balance tank pressures alongside the production rate and voidage history.

***Reservoir Engineering Results and Conclusions.*** From the dynamic models, it was found there was a good overall history match for the Middle Minagish reservoir, though it was difficult to match some wells due to a lack of pressure data to calibrate the productivity. Other wells also had early water breakthrough in the model, also impacting history match quality. From the pressure mapping, the areas of highest depletion in the Middle Minagish are found to the south of the field, away from incident water well. The lowest pressures across the whole field were seen in the Mishrif formation with areas of high localised depletion, consistent with the low permeability nature of the reservoir. Analysis of the GOC from logs vs the model showed poor alignment leading to the conclusion the model was unable to directly show where leaks were. The PNC log data showed a general shallowing of the GOC over time with a general trend of there being deeper GOC's in the south, consistent with the observed lower pressures. The material balance study indicated a disparity between expected and actual pressures, and is interpreted to indicate the approximate time of the start of the leak.

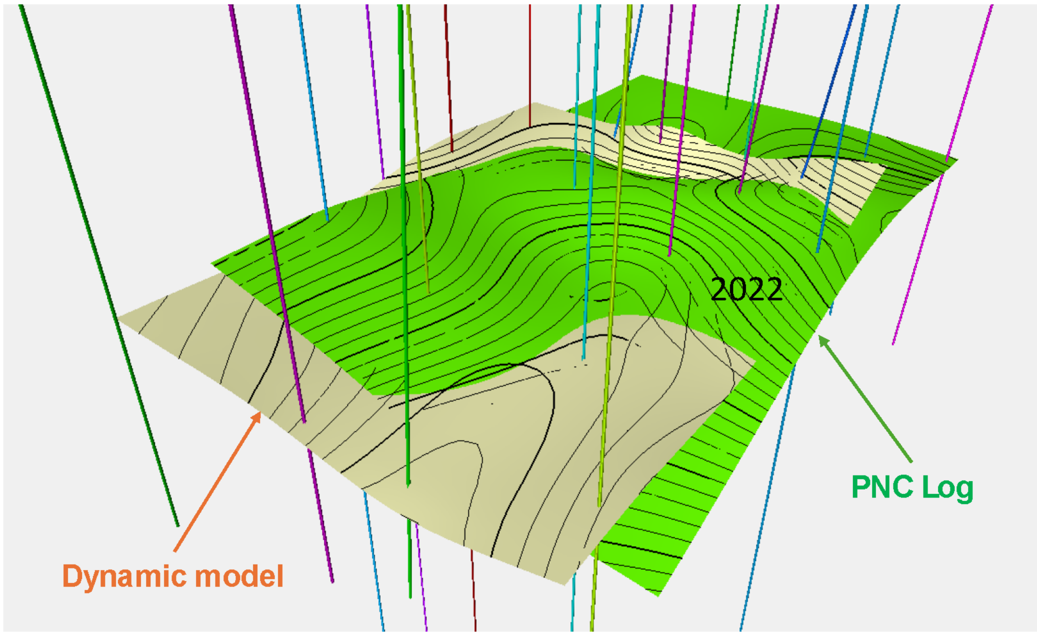



**Figure 8—Differences in the gas-oil contact (GOC) between the dynamic model and the measured PNC log data making the dynamic model unsuitable for the prediction of potential leaks.**

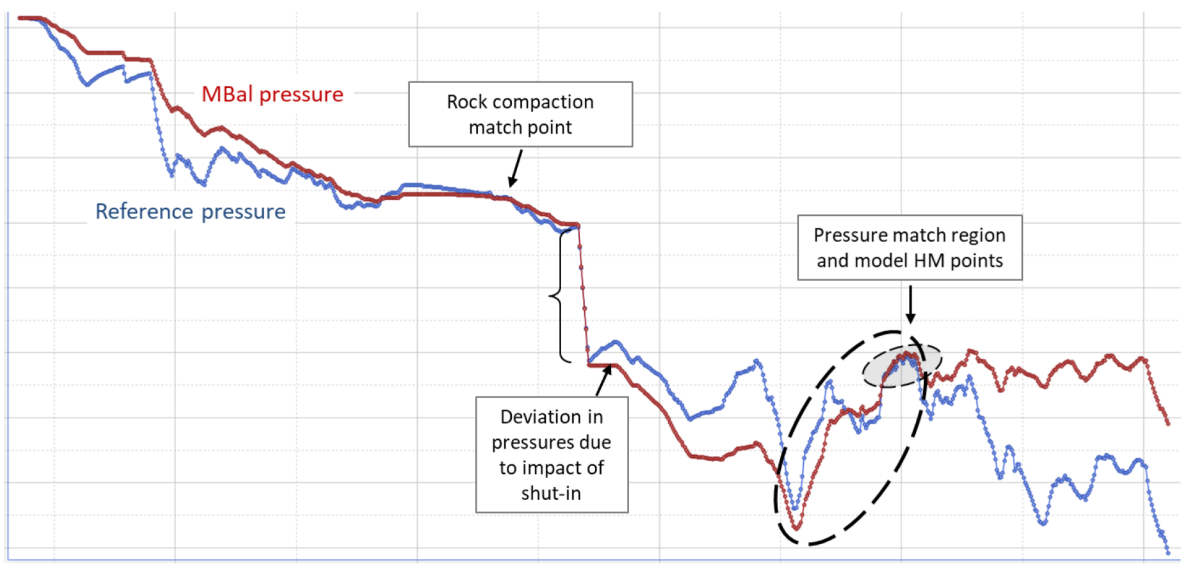



**Figure 9—Material balance outcome showing pressure in the reservoir over time. The separation of the lines is an indicator of more offtake than expected, and therefore a potential indicator of the start of a leak. The deviation beforehand is explained by the differences between how material balance handles a long term shut in versus the real reservoir.**

## Geomechanics

The Geomechanics part of this work focused on five key points: cap rock and formation integrities, compaction risks, fault reactivation potential and hydraulic fracture risks. The outcome of this focus would be presented on a per well basis via the well montages (see below).

***Geomechanics Workflow.*** To perform this analysis a geomechanical stress and strength model was developed and calibrated to core and field data. The stress model was built primarily using petrophysical log data as there was extensive density and sonic log coverage, and some shear sonic data also available. Stress indicators were taken from mini-frac tests, step rate (Sanaee et al., 2010) and leak off tests. Image logs were used to support stress orientation. Young's modulus and Poisson's ratio were derived from sonic logs and lab data, with strength parameters (UCS and friction angle) estimated using empirical correlations. Synthetic logs were used to estimate shear velocity where data was missing. Vertical stress was calculated from density logs and formation thickness (Zoback et al., 2003). The minimum horizontal stress was derived from mini-frac and leak off data, then calibrated using Tectonic Strain and Matthews & Kelly models (Mathews and Kelly, 1967). The maximum horizontal stress was estimated from breakout analysis (Zoback et al. 1985) and wellbore stability modelling. The calibrated stress model was used in conjunction with reservoir pressure and damaged zone (fault and fracture) analysis to determine the relevant geomechanical risks that could lead to a leak path.

A generalized stress model along with available field calibration data is shown in Figure 10. While built using log, core and field data, this generalized stress model was able to be replicated in all wells field wide without the need for petrophysical logs using only a TVD curve and set of formation tops.

***Geomechanics Results and Conclusions.*** From this model, three main conclusions were drawn. Firstly, the cap rock is at low risk of fracturing from injection pressures. Some water injectors do approach the minimum horizontal stress due to thermal effects, but this is in the Minagish reservoir and not the caprocks and is localized only to those wells. Secondly, fault slip risk is mostly low, though there is some increase in risk due to thermal injection effects. There is no fault slip predicted for the cap rocks and overburden formations. Finally, there is some compaction risk in zones with depletion, which is unlikely to have a direct link with potential leak paths, however this could lead to wellbore integrity issues.

No single geomechanical mechanism explains the leak path, however geomechanical risks could combine with other mechanisms (i.e. compaction affecting cement integrity and hence well integrity) that could in one way or another contribute to the leak path risk.

## Production Chemistry

The Production Chemistry component of the study was focused on compositional fingerprinting analyses of leak fluids with the aim of determining of reservoir(s) of origins and pathway to surface. In addition to this, sulphate scale modelling of leak brine + aquifer water mixtures was also carried out, as was assessment of shallow well corrosion evidence, both with the objective of supporting wider study conclusions on the root cause of the blowout.

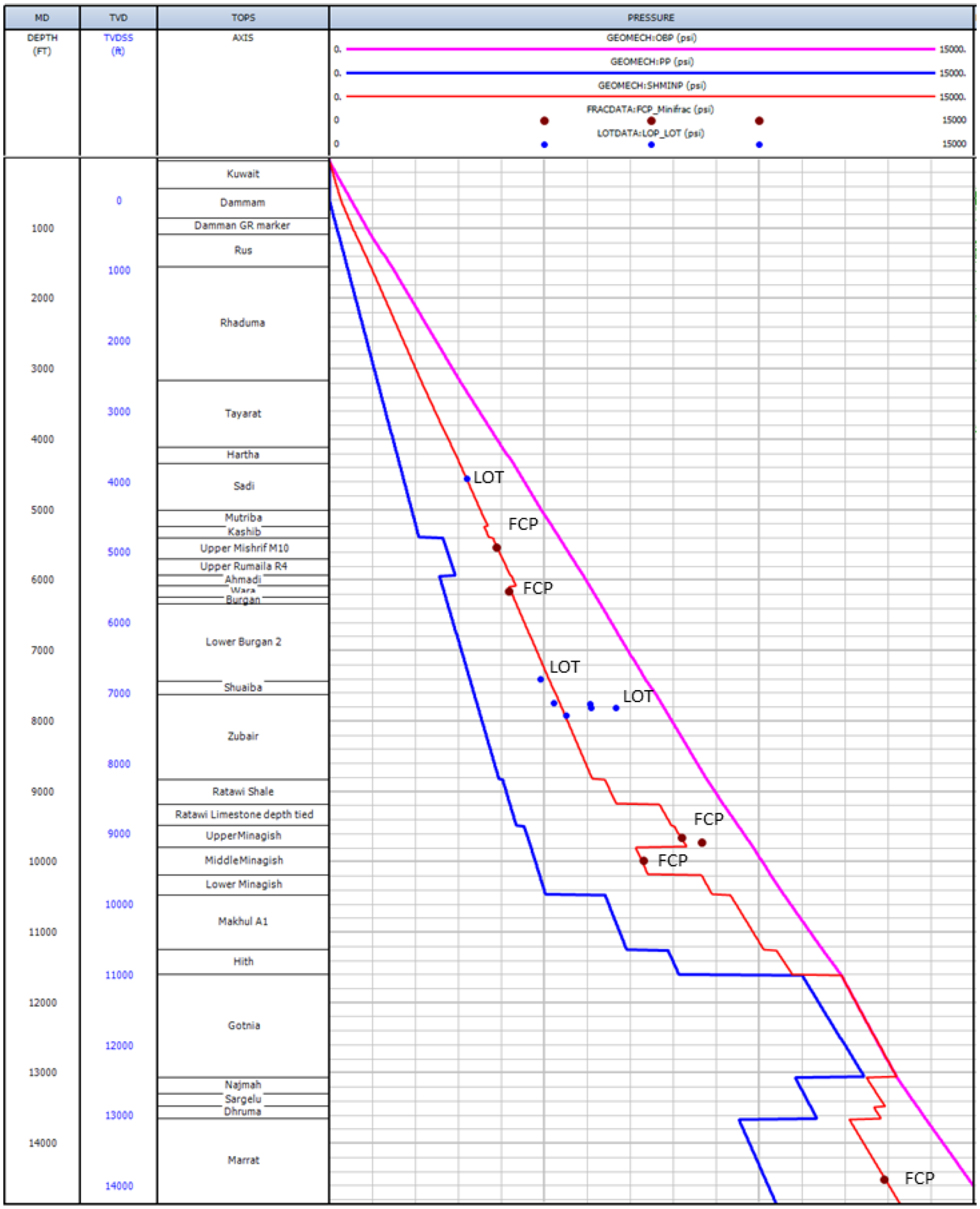



**Figure 10—Generalized stress model for the Minagish field including pore pressure at virgin conditions (blue line), vertical stress (pink line) and minimum horizontal stress at virgin conditions (red line). Field data available for model calibration plotted, including leak-off test data (blue dots) and fracture closure pressure corrected back to virgin conditions (brown dots).**

***Production Chemistry Workflow.*** A fluids data review with QA/QC was undertaken. This included reservoir PVT reports, well surface sampling results, and all previous leak fluid analyses. Gaps in data were identified, leading to a priority water sampling programme covering leak relief, oil production and shallow aquifer wells.

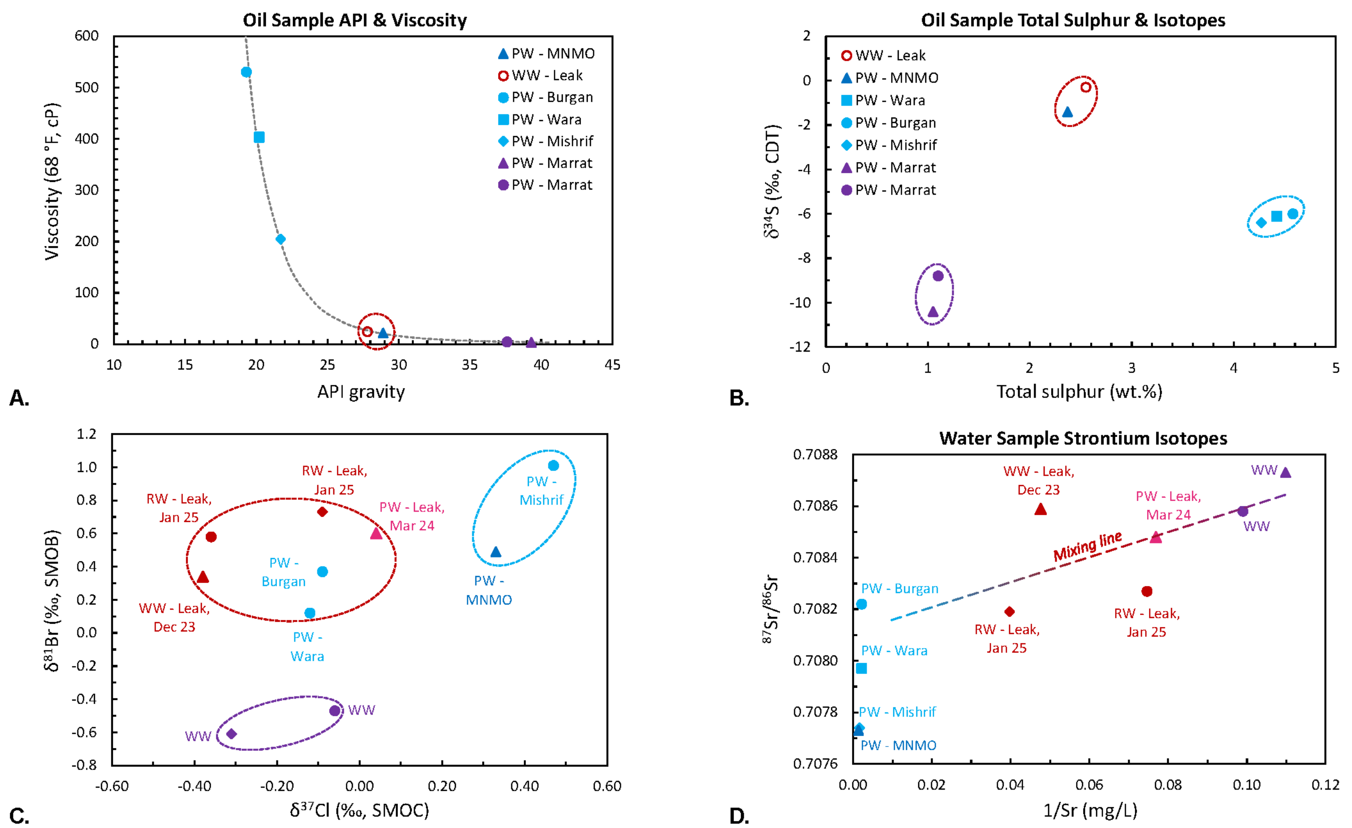
The chemical fingerprints of leak oil, gas and brine samples – based on a range of compositional measurements (including oil C36+, SARA [saturates, aromatics, resins, asphaltenes], high resolution GC [gas chromatography] saturate and aromatic ratios, gas C15+, water major ions and trace elements) plus stable isotope data (hydrocarbon $\delta^{34}S$, $\delta^{13}C$ and $\delta D$, brine $^{87}Sr/^{86}Sr$, $\delta^{37}Cl$ and $\delta^{81}Br$) – were compared with those of local reservoirs to pinpoint origins (Table 1 and Figure 11).

**Table 1—Selected Production Chemistry oil, gas and water fingerprinting data for various Minagish field wells. Oil and gas from the leak water well most closely matched the main Minagish oolite (MNMO), with gas matching those wells in the non-native injection gas cap. By contrast, saline leak water entering the shallow aquifer most closely matched that for Burgan / Wara wells.**

| | | | Oil | | | | | | | | Gas | | | | | Water | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Formation | Well | C36+ | SARA | API | Viscocity | S wt. % | $\delta^{34}S$ | HRGC-S | HRGC-A | C15+ | $H_2S$ % | $H_2S$ $\delta^{34}S$ | $CH_4$ $\delta D$ | $CH_4$ $\delta^{13}C$ | $^{87}Sr/^{86}Sr$ | $\delta^{37}Cl$ | $\delta^{81}Br$ |
| | Leak | PW | | ✓ Reasonable | ✗ No match | ✗ No match | ✗ Poor | ✗ Poor | ✓ Reasonable | ✓ Reasonable | ✓ Good | ✓ Good | ✓ Good | ✓ Reasonable | ✓ Good | ✓ Good | ✓ Reasonable | ✓ Good |
| Inj. gas cap | Minagish | PW(FGI) | | | | | | | | | ✓ Good | ✓ Reasonable | ✗ Poor | ✓ Reasonable | ✓ Good | | | |
| Inj. gas cap | Minagish | PW(FGI) | | | | | | | | | ✓ Good | ✓ Reasonable | ✗ Poor | ✓ Reasonable | ✓ Good | | | |
| Inj. gas cap | Minagish | GI | | | | | | | | | ✓ Good | ✓ Good | ✓ Reasonable | ✓ Good | ✓ Good | | | |
| Inj. gas cap | Minagish | PW(FGI) | ✓ Good | ✓ Good | ✓ Good | ✓ Good | ✓ Good | ✓ Good | ✓ Good | ✓ Good | | | | | | | | |
| Associated gas | Minagish | PW | | | | | | | | | ✗ No match | ✗ No match | ✗ Poor | ✓ Good | ✗ No match | | | |
| Associated gas | Minagish | PW | | | | | | | | | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | | | |
| Associated gas | Minagish | PW | | | | | | | | | ✗ No match | ✗ No match | ✗ Poor | ✗ No match | ✓ Reasonable | | | |
| Associated gas | Minagish | PW | | | | | | | | | | | | | | ✗ No match | ✗ No match | ✓ Good |
| Associated gas | Minagish | PW | | | | | | | | | ✗ Poor | ✗ Poor | ✗ Poor | ✗ No match | ✗ No match | | | |
| Associated gas | Mishrif | PW | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ Poor | ✗ No match | | | | | | | | |
| Associated gas | Wara | PW | | | | | | | | | | | | | | ✓ Reasonable | ✓ Good | ✓ Good |
| Associated gas | Wara | PW | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ Poor | ✗ No match | | | | | | | | |
| Associated gas | Burgan | PW | | | | | | | | | | | | | | | | |
| Associated gas | Burgan | PW | | | | | | | | | | | | | | ✓ Reasonable | ✓ Good | ✓ Good |
| Associated gas | Burgan | PW | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ Poor | ✗ No match | | | | | | | | |
| Associated gas | Marrat | PW | | | | | | | | | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✓ Reasonable | | | |
| Associated gas | Marrat | PW | | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✓ Reasonable | ✓ Reasonable | ✗ No match | ✗ No match | ✓ Reasonable | | | |
| Associated gas | Marrat | PW | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | ✗ No match | | | | | | | | |

| Match with water well (WW) leak fluids | |
|---|---|
| ✓ | Good match |
| ✓ | Reasonable match |
| ✗ | Poor match |
| ✗ | No match |
| | No data / N.A. |

| | |
|---|---|
| PW | Production well |
| GI | Gas injection well |
| PW(FGI) | Production well (fmr. gas injector) |




PW = Production wells, WW = Water wells, RW = Leak relief wells

**Figure 11—Example Production Chemistry leak source fingerprinting data. (A) Oil sample API gravity vs viscosity and (B) sulphur content vs isotope ratio ($\delta^{34}S$); water well (WW) leak fluids are a match with MNMO oil in both cases. (C) Gas methane carbon ($\delta^{13}C$) / hydrogen (deuterium, $\delta D$) isotopic ratios; leak gases most closely match non-native injection gas cap wells. (D) Water dissolved strontium ($^{87}Sr/^{86}Sr$) isotope ratios; leak waters lie on a mixing line between local aquifer water wells and Wara / Burgan formation produced waters.**

Sulphate scale modelling involved simulating the mixing of saline leak brines with brackish shallow aquifer water at relevant temperatures / pressures to assess potential for scale precipitation around well casings based on both critical scale saturation and mass (Mackay et al., 2005). Shallow well corrosion was evaluated in respect of (internal) produced fluids and (external) local aquifer geochemical conditions.

***Production Chemistry Results and Conclusions.*** Production Chemistry fingerprinting demonstrated the following three key points (Table 1, Figure 11): (1) Leak oil and gas were from the main Minagish Oolite (MNMO) reservoir, with no obvious evidence of interaction with other local hydrocarbon reservoirs during ascent, (2) leak gas was the non-native injection gas which forms the artificial gas cap at the Minagish reservoir crest, and (3) leak fluids entering the shallow aquifer contained a significant saline reservoir brine component, most likely drawn from a Wara / Burgan (overlying the MNMO) source.

Integrated with geological, geomechanical and well integrity findings, geochemistry results were key to concluding on a direct (well) path to surface, and to narrowing down suspect wellbores out of the ~600 in the field (MNMO wells penetrating the gas cap only).

While saline leak water source fingerprinting proved challenging due to the compositional similarity of reservoir brines combined with a high level of dilution (90%) in shallow aquifer waters, it was demonstrated that stable isotopes ($^{87}Sr/^{86}Sr$, $\delta^{37}Cl$, $\delta^{81}Br$) provided a useful means to differentiate between local and/or deep formation water sources (Kendall and Caldwell, 1998; Hendry, 2000; Barnaby et al., 2004; Marza et al., 2022). Chlorine $\delta^{37}Cl$ data was particularly indicative of a leak water Wara / Burgan origin (Figure 11C), supported by extrapolation of the $^{87/86}Sr$ mixing line for aquifer and blowout relief well water samples (Figure 11D).

Scale modelling supported calcium sulfate ($CaSO_4$) scale deposition – which would be expected to contain trace radioactive radium sulfate ($RaSO_4$) (Smith, 1987; Ghose and Heaton 2005; MacIntosh, et al., 2024) – associated with shallow leak brine crossflow around warm production wells as a cause of widespread well gamma ray anomalies.

At least one identified case of severe external ‘outside in' well casing / tubing corrosion at shallow depth was concluded as microbial (Brondel et al., 1994; Kiani Khouzani et al., 2019), likely fed by sulphate rich aquifer waters (Al Ruwaih, 1995; Al Ruwaih et al.,1998), and potentially aggravated by sour leak fluids entering the latter. Shallow casing corrosion is a known problem Kuwait, and has been previously linked to poor cementation due to losses (Agiza and Awar, 1983). This, combined with typical ‘inside out' corrosion associated with sour produced fluids (Brondel et al., 1994; Al Salem, 2023), was therefore concluded as a likely contributing factor to shallow well corrosion opening up leak paths from production wells into the aquifer.

## Well Data Review

The Wells scope for this project consisted of three main parts: (1) Support identification of potential leak paths with a deep dive of suspect wells, (2) assessment of appropriate well construction and surveillance logs, and (3) deliver a well integrity montage for each wellbore. As will be shown, this was not performed in isolation, and inputs from the various teams working on this were required to deliver an integrated and quality outcome.

***Well Workflows.*** There were a significant number of logs of various types supplied for this project (Table 2) of differing vintage, supplier and quality of analysis. The generalized workflow for the log interpretation is as follows (Figure 12). It was important to agree with all stakeholders the parameters that were to be used for this work across the various logs.

Table 2—Various log types and approximate numbers of logs processed for analysis.

| Log Type | Approximate Count |
|---|---|
| Cement Bond Log | 1010 |
| Pulsed Neutron Log | 80 |
| Noise Log | 60 |
| Electromagnetic log | 50 |
| Multi-finger caliper log | 140 |
| Temperature log | 150 |
| Gamma log | 190 |
| Ultrasonic log | 130 |

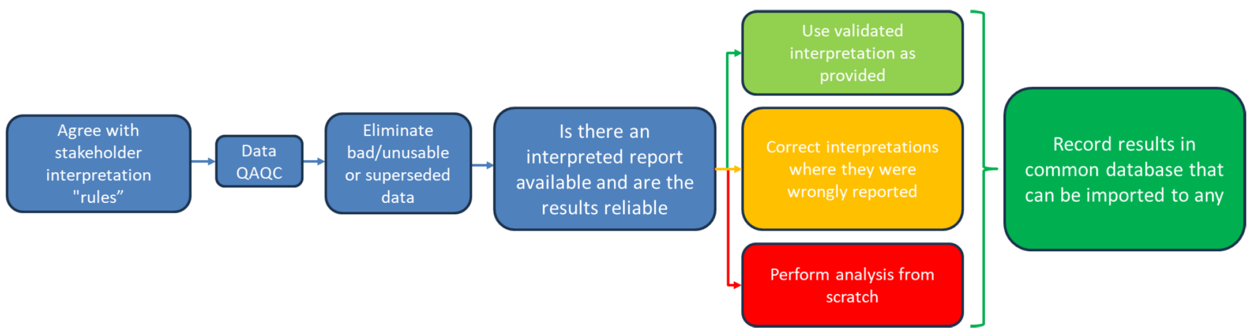



Figure 12—High level workflow for the Petrophysics work. Agreeing the framework with all stakeholders is critical, as then the subsequent QA/QC can be targeted appropriately.

As this project was always requiring assessment of all logs, regardless of whether or not there was a clear connection to the shallow water formation leak, being clear and organized was foundational. Sharing data in a meaningful way for other technical disciplines on this project also was key for efficient sharing of new found information and creating the final well montages.

The bulk of the well review work was spent assessing the various well construction and well operation documents, such as daily drilling reports (DDRs), mud reports/logs, workover reports. This data was collated, reviewed, augmented with some basic calculated values (such as theoretical top of cement) and finally integrated with the other technical elements of this project to present an integrated view of the well. The project used Gravitas Edge from HRH Geology to present well montages; an example is shown in Figure 13.

There was now in place a reviewed, referenced and cross discipline view on every wellbore under investigation. This provided the basic tools for a deeper exploration of a subset of wellbores, the key was how to narrow down nearly 600 wellbores to a reasonable number of candidates. To do that, the decision tree shown in Figure 14 was developed.

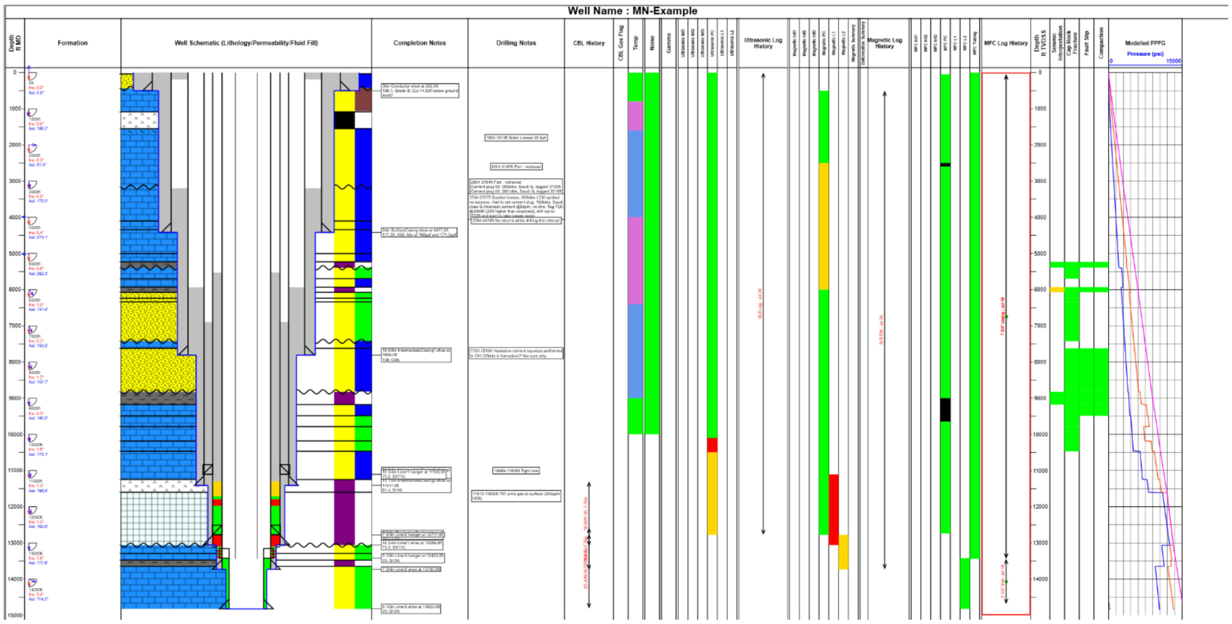



Figure 13—An example well montage produced using Gravitas EDGE from HRH Geology. The illustration shows the integration of the various disciplines, starting on the left with the geological data, overlaid with the final well construction (with cement bond log where available), including any observations of note. The logs and their history is presented next, with the geophysical and geomechanical outputs shown on the right hand side.

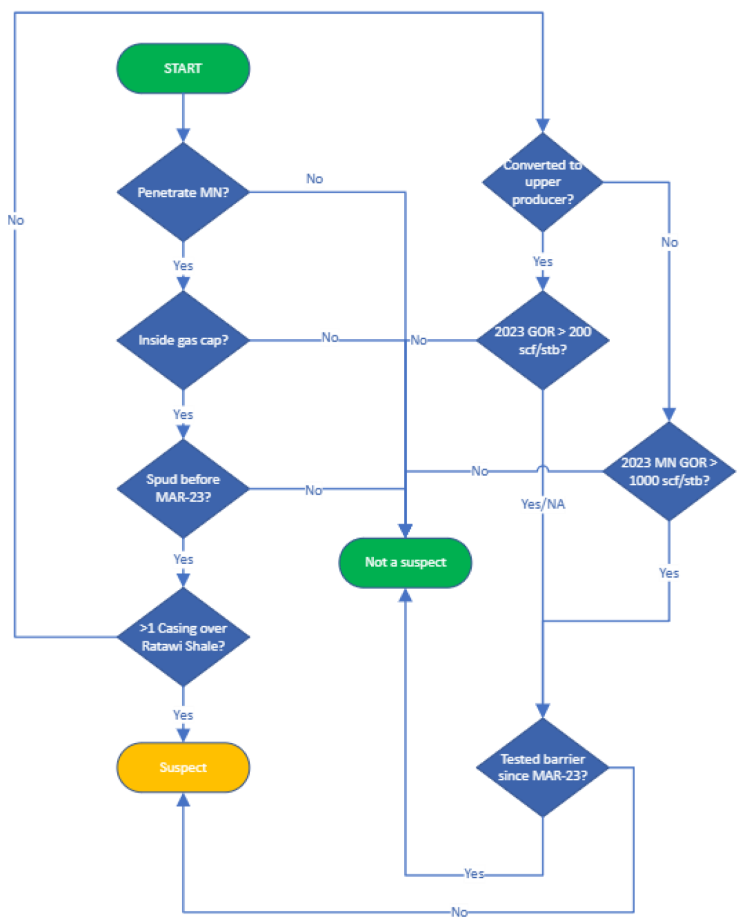



Figure 14—Decision Tree diagram to assess whether a given well is a likely suspect for a vertical leak. This brings together various technical disciplines' inputs.

This flow chart was developed with the larger integrated team, as various elements required specific inputs (Production Chemistry showed the fluid came from Minagish reservoir gas cap; Reservoir Engineering confirmed the areal spread of the gas cap; Geology showed where the seals were). Running all the wellbores through this led to 24 wellbores that were labelled as suspect. Those were looked at in fine detail to explore their likely contribution and any remedial actions required.

***Well Results and Conclusions.*** A well integrity montage was generated for each wellbore, showing a fully integrated view of the well, from construction and surveillance to geology and geomechanics. The significant effort reviewing all the supplied petrophysics and integrity logs was shown on these plots, with detailed colour coding of cement bond logs where available. A decision tree was produced to help narrow down the list of suspect wells to 24. Those wells were then explored more deeply, with 2 being identified as high probability of contributing to the leak. High level remedial actions were recommended. For the remaining 22 wells, surveillance actions were recommended to further inform whether those wells were also potentially contributing to the vertical leak.

## Discipline Results Findings Integration

The key findings of each discipline that were relevant to the leak path were:

- Production Chemistry showed the fluid came from the Minagish reservoir gas cap
- Reservoir Engineering confirmed the areal spread of the gas cap and the relevant reservoir and aquifer pressures that are likely to counter any vertical migration of fluid
- Geology showed where the seals were and showed the potential for Karst highways linking problem wells with event wells
- Geomechanics confirmed a vertical leak path through faults / fractures was unlikely
- Well Engineering confirmed the existence of wells contributing to a vertical leak path and narrowed down to 24 a list of suspects, with remedial actions where required

Bringing those together in an integrated way as the project went along, and not simply at the end, allowed for a reduction of the complexity of the problem, and a clear most likely outcome to emerge from what was a complex initial picture. Table 3 below summarizes the range of the problem and how working in this way allowed the most likely route to come to the fore.

The key recommendations were as follows:

- Execute the recommended remedial actions on the high risk suspect wells
- Gather surveillance on the lower risk suspect wells to remove them from the list
- Further focused fluid sampling to reduce uncertainty
- Improve material balance and simulation models with that data
- Review materials selection and cementing techniques to reduce future corrosion potential
- Assemble and deliver a data acquisition programme for the shallow aquifer formation (high resolution, shallow seismic, tracers etc)

Most of the evidence presented led to the conclusion that the gas in the shallow water formation was directly from the Minagish reservoir, specifically the gas cap of that reservoir, previously charged with gas from a different field. As such this gas had migrated directly up one or more wellbores directly to the Khasib or higher and then entered the formation via damage and holes, most likely formed by both exterior and interior corrosion. 24 wells were identified as potential suspects, with two very clearly showing evidence of such a pathway. Other routes were considered within the evidence presented, and ranked in Table 3 above.

Table 3—A summary of the potential leak paths, with evidence both for and against, with conclusions and next steps. The top route, namely a vertical leak via one of more wellbores into the Khasib reservoir or higher, and a horizontal migration to the event wells, was felt the most likely.

| Gas leak mechanism | Evidence showing this is mechanism | Evidence showing this is not the mechanism | Overall conclusion | Recommendations |
|---|---|---|---|---|
| **Through wellbores and out by internal /external corrosion to any formation above Khasib** | • Corrosive fluid in Shallow waters<br>• Wells prone to corrosion in shallower zones (corrosion logs)<br>• Gas flow from corroded production well cellar<br>• Massive corrosion in some wells | Recent positive pressure test required to prove well is intact which is not available in all wells | This is considered the main / primary cause of the incident water well. There is a high likelihood that gas pockets remain in other locations and that vertical cross flow is occurring in numerous wellbores between different formations. | • Pressure tests in suspect wells<br>• Inspection of casing condition<br>• Noise, Cement Bond Log (CBL), temperature logs<br>• Chemical sampling |
| **Through external wellbore annulus** | Certain well noise logs and gas (that may or may not be injected gas) detected in some well annuli | • Past Zubair not possible in any meaningful way due to differential pressure<br>• Past Wara / Burgan not likely as they act as pressure sinks | Fluid movement in external annuli above Khasib could be part of the movement of fluids but direct annular channel from Minagish to above Khasib less likely | • Noise, ultrasonic temp, CBL logs<br>• Chemical sampling |
| **More tortuous crossflow route through various annuli and via reservoirs** | • Metal loss and corrosion in various horizons – there are continuous sands that are capable of cross flow<br>• Chemical signs of Wara Burgan water in the incident water well | • Chemical data does not support significant mixing of different reservoir shallower fluids in incident water well<br>• Overpressure Zubair would suppress flow outside casing<br>• Multiple pressure loss zones to pass through outside casing | Hard to explicitly rule out that this could be occurring but is not considered the main leak pathway for incident water well | Chemical sampling |
| **Through geology / faults / fractures** | • Seismic data shows area of higher disturbance with a noted East West trend in deeper horizons<br>• In MM formation only, combinations of pressure and temp changes could activate faults or cause thermal fracs | • Separate reservoirs of different hydrocarbons have formed over geological time<br>• The volume / rate of gas is significant compared to likely fault flow rate<br>• Under /over pressured zones would act as barriers or sinks<br>• No chemical trace of shallower hydrocarbons<br>• No clear and continuous mapped fault zones from Minagish to shallow water Fm and fault vertical throw is small | Local fault damage and thermal / hydraulic fractures in MM could lead to locally increased permeability and conductive channels but only within this horizontal layer. | Shallow seismic of the shallow water formation may assist in understanding connectivity |

# Conclusions

The aims, multidisciplinary workflows with technical results, and integration of these to resolve a complex problem – a reservoir fluid leak to subsurface of unknown origins in an oil field with several producing reservoirs and hundreds of wells – have been presented. From this, the following is concluded.

Determining the origins of the water well blowout at Minagish was a multi-faceted problem, with these closely interlinked. Relevant disciplines working in isolation risked missing an important part of the puzzle. As a result, while each had its own workflows, aligning these and bringing their technical findings together was key to understanding potential leak paths so developing remedial actions. This was successfully achieved by ensuring each discipline had clear objectives aimed at feeding into an evolving logical process, namely determining the origins of leak fluids, their possible route(s) to surface, before finally homing in on the potential leak path locations amongst the myriad of wells in the field.

Production Chemistry fingerprinted the source of the leak and confirmed a lack of interaction with fluids from shallower reservoirs. Reservoir Engineering determined the areal spread of the gas cap and the relevant reservoir / aquifer pressures which would counter vertical migration of fluids. Geology mapped seals as barriers to vertical migration while assessing potential for shallow Karst highways linking problem wells with event wells. Geomechanics ruled out a vertical leak path through faults/fractures, leaving leaking wells was the only viable route to surface. Finally, the wells team confirmed the existence of wells with integrity issues which could contribute to a vertical leak path, and narrowed 600 down to a list of 24 suspects, with remedial actions.

Ultimately, understanding the leak cause and route to surface is key to putting mitigations in place to ensure hydrocarbon influx into the shallow water formation is halted. This should allow the aquifer to be returned safely and environmentally soundly to water production only. The developed workflow can be readily applied to other fields, including as a preventative strategy. The root causes of the leak at Minagish are widespread problems regionally, and plans are already underway to risk assess Kuwaiti fields using the developed multidisciplinary workflow.

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